# K-Means Clustering: Experimental Study

**CS5100 – Foundations of Artificial Intelligence**

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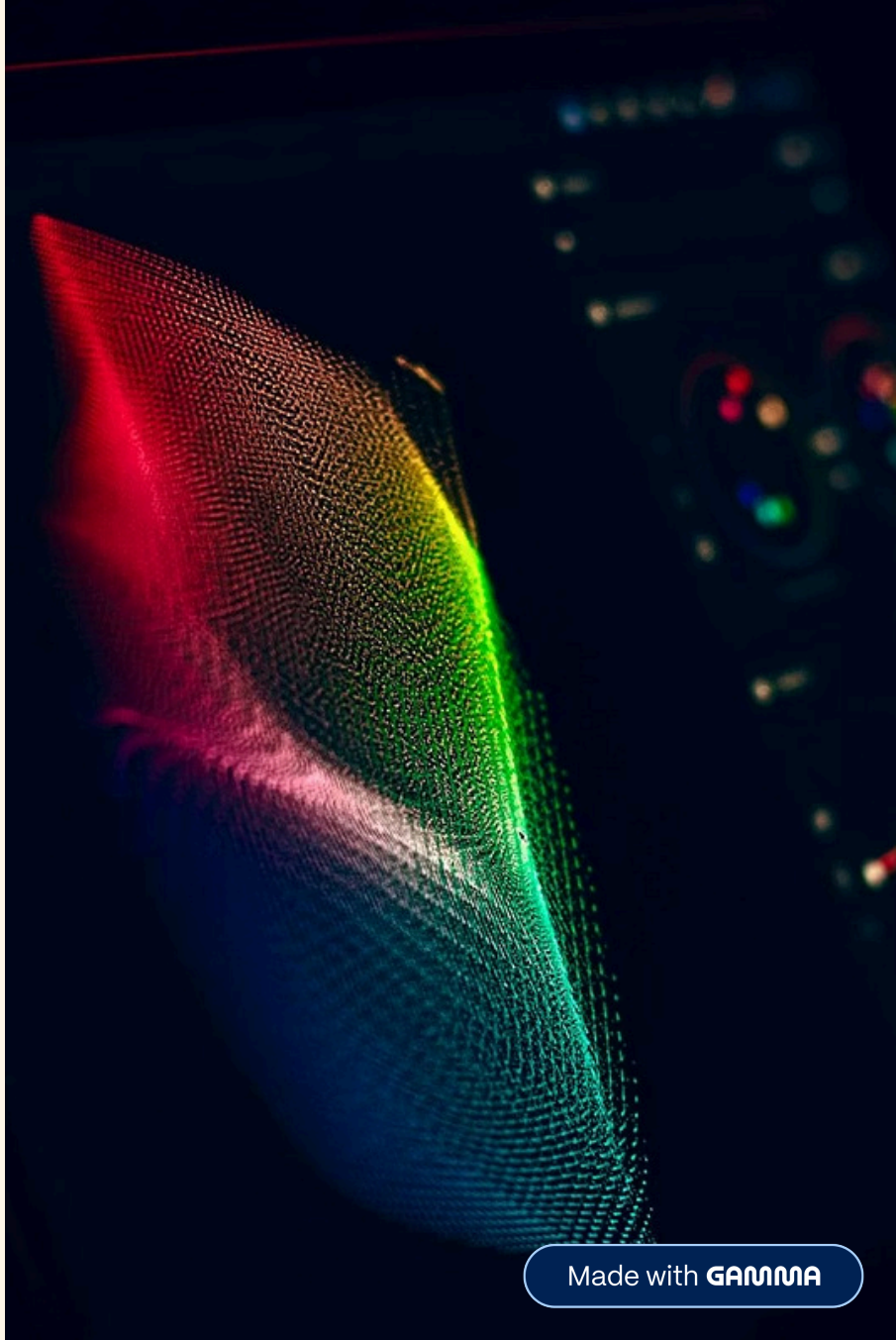

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# Project Overview

This experimental study extends MacQueen's seminal 1967 K-means algorithm through systematic analysis of initialization methods, cluster validation techniques, and algorithmic variations.

### Elbow Method

Systematic K-value analysis using inertia metrics

### Silhouette Analysis

Quantitative cluster quality assessment

### K-Means++

Improved centroid initialization

### PCA Visualization

Dimensionality reduction for interpretability

### K-Median

Robustness to outliers

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# Problem & Motivation

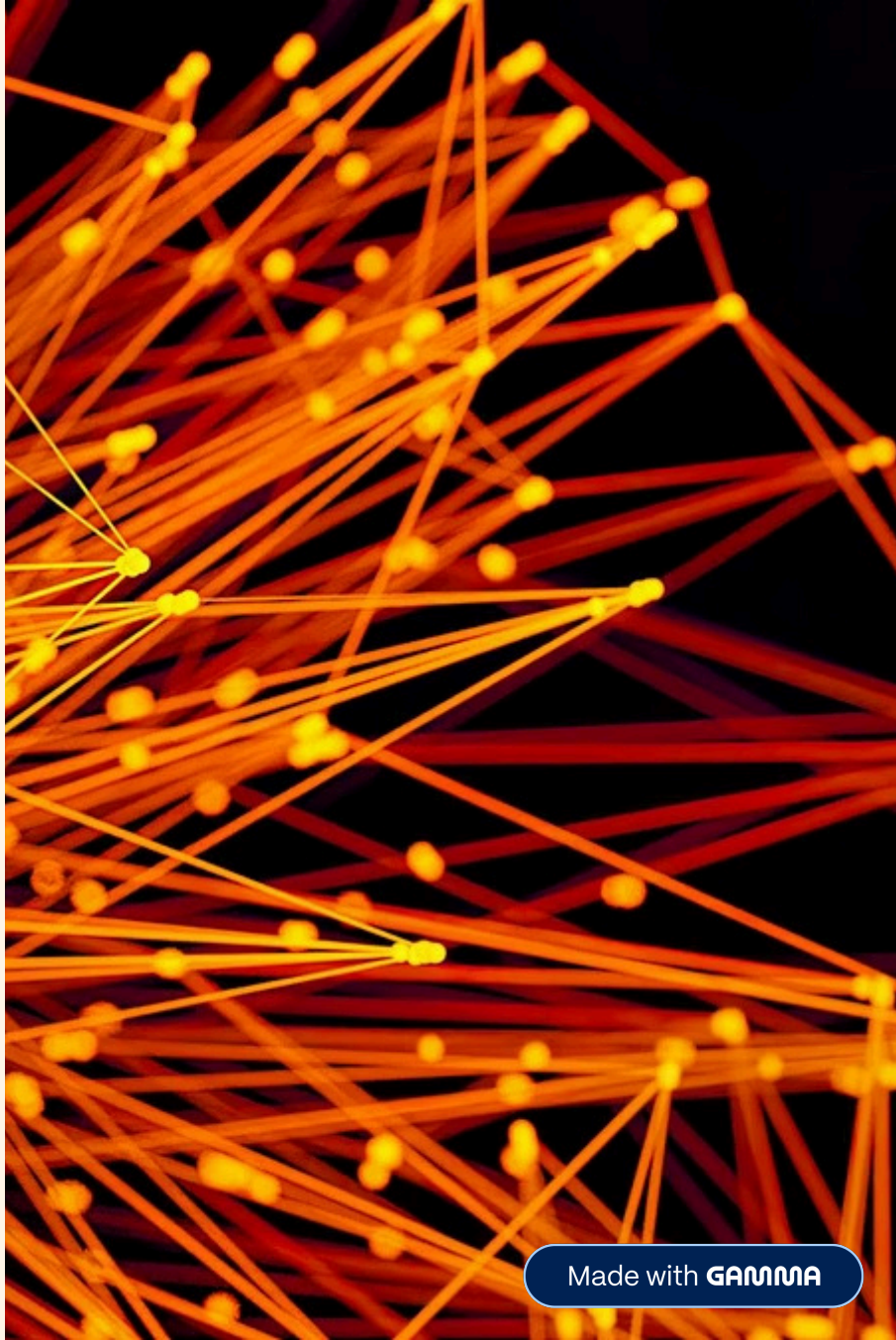
## What is Clustering?

Clustering represents a fundamental unsupervised learning technique that groups data points based on similarity, enabling pattern discovery without labeled training data.

## Key Challenges

- Requires predetermined number of clusters (K)
- Random centroid initialization affects results
- High-dimensional data resists visualization

These limitations motivated our investigation into enhanced initialization methods and validation techniques.



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# K-Means Algorithm

MacQueen's iterative algorithm partitions data into K clusters through repeated centroid updates.

1

### Initialize Centroids

Random placement in feature space

2

### Assign Clusters

Each point joins nearest centroid

3

### Update Centroids

Calculate mean of cluster members

4

### Check Convergence

Repeat until stability

**Implementation:** Batch processing updates all centroids simultaneously after full dataset assignment

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# Elbow Method for K-Selection

The elbow method systematically evaluates different K values by plotting inertia (within-cluster sum of squares) against cluster count.

01

## Range Selection

Test K values from 1 to maximum reasonable clusters

02

## Inertia Calculation

Compute total within-cluster variance for each K

03

## Visual Identification

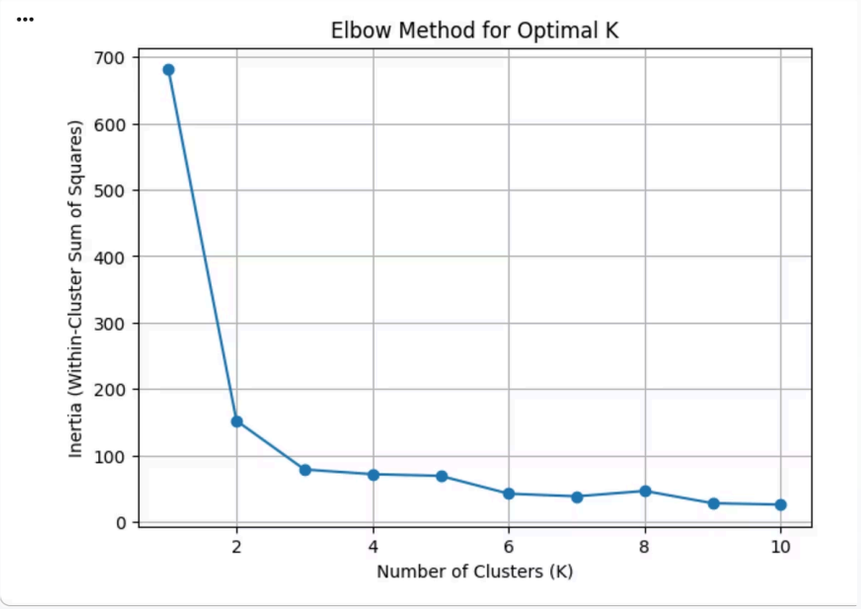
Locate "elbow" point where inertia reduction plateaus

04

## Optimal K

Balance model complexity with diminishing returns

For the Iris dataset, analysis revealed **K ≈ 3** as optimal, matching species count.

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# Silhouette Analysis

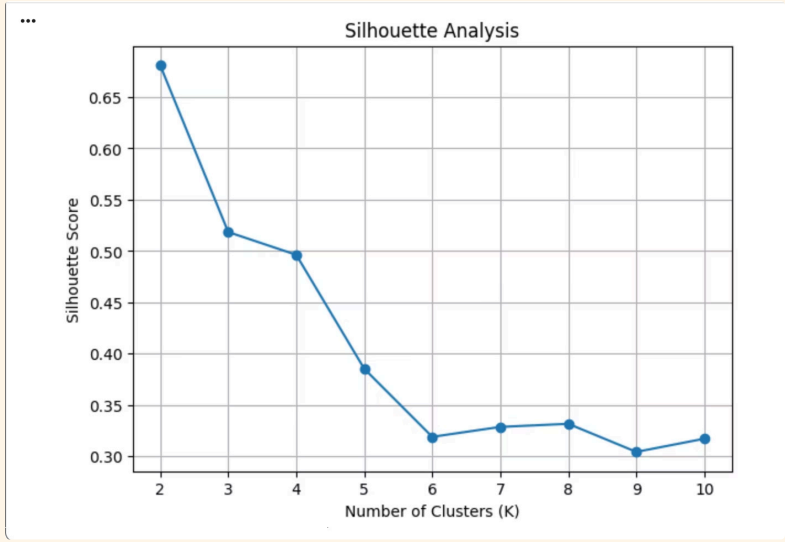
## Quantitative Cluster Quality

Silhouette coefficients measure cluster cohesion and separation using:

- **a(i)**: average distance to same-cluster points
- **b(i)**: average distance to nearest-cluster points
- **s(i)** = [b(i) - a(i)] / max{a(i), b(i)}

Scores range from -1 to 1:

- **Close to 1**: Well-clustered
- **Close to 0**: Overlapping clusters
- **Close to -1**: Misclassified points




**Result:** Maximum silhouette score at K = 3 validated elbow method findings

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# K-Means++ Enhancement

K-Means++ addresses random initialization instability through intelligent centroid selection.

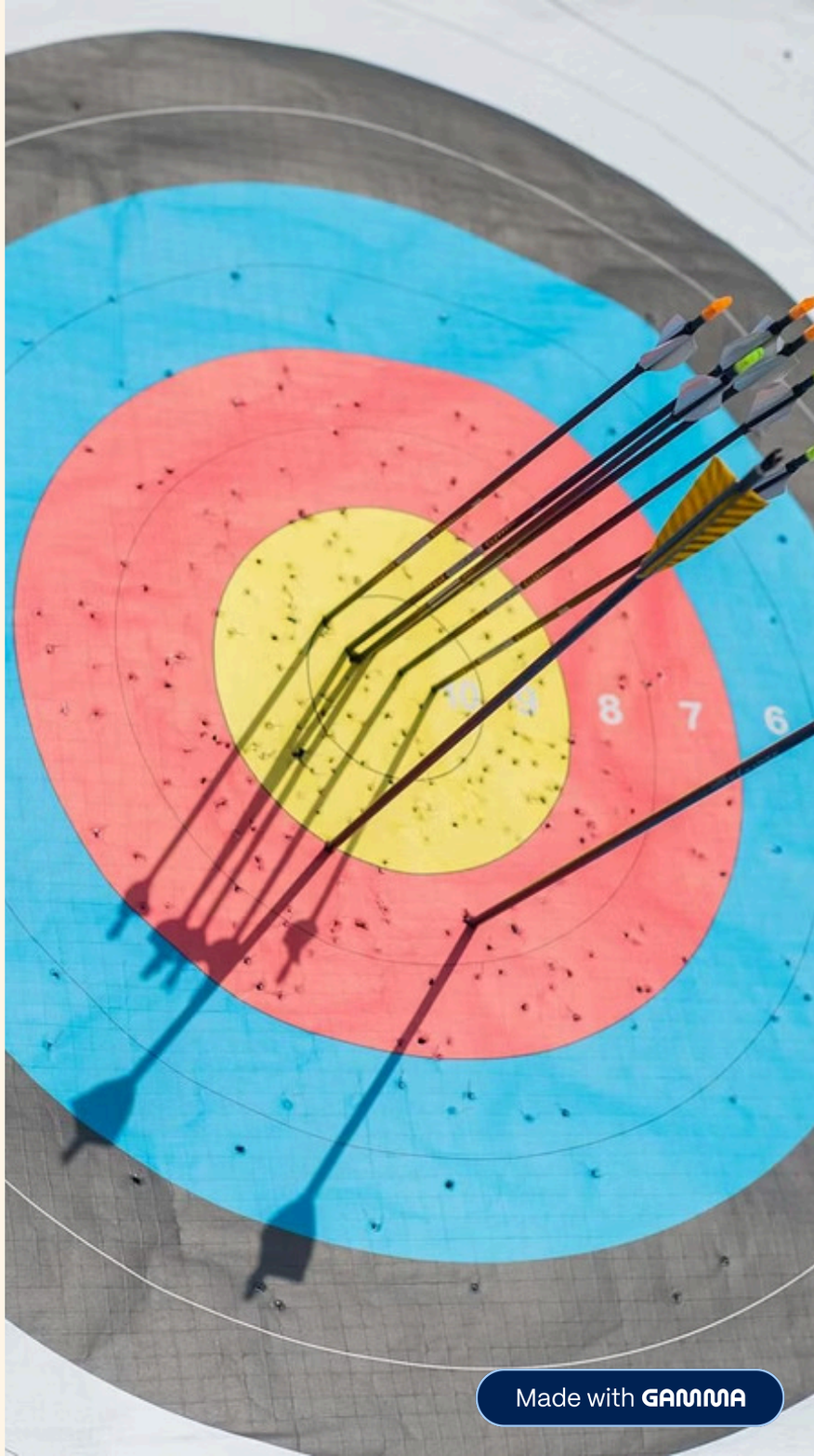
1. **First Centroid**
   Random selection from data points
2. **Probability Weighting**
   Select subsequent centroids proportional to $D^2(x)$
3. **Iterative Placement**
   Repeat until K centroids positioned
4. **Standard K-Means**
   Proceed with enhanced initialization

## Advantages

- More stable clustering results
- Faster convergence
- Reduced random initialization variance

## Theoretical Guarantee

O(log K)-approximation to optimal clustering compared to standard K-means



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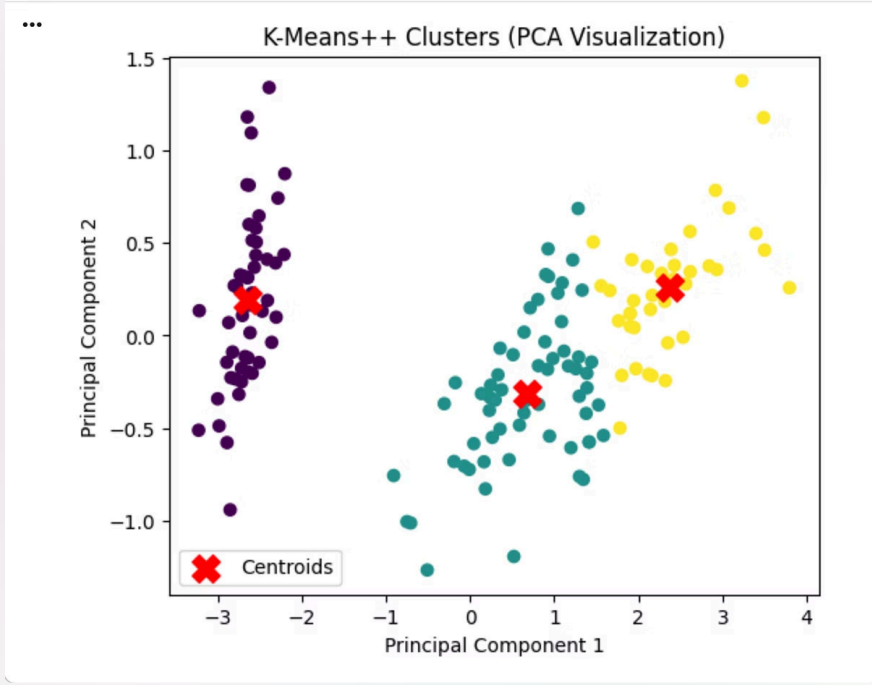


# PCA Visualization

Principal Component Analysis enables cluster visualization through dimensionality reduction.

### 4D Iris Data

Sepal length, width, petal length, width

### PCA Transformation

Project to principal components preserving variance

### 2D Projection

Visualize clusters in reduced space

Visualization revealed **clear separation** between Iris species, validating clustering quality.

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# K-Median Extension

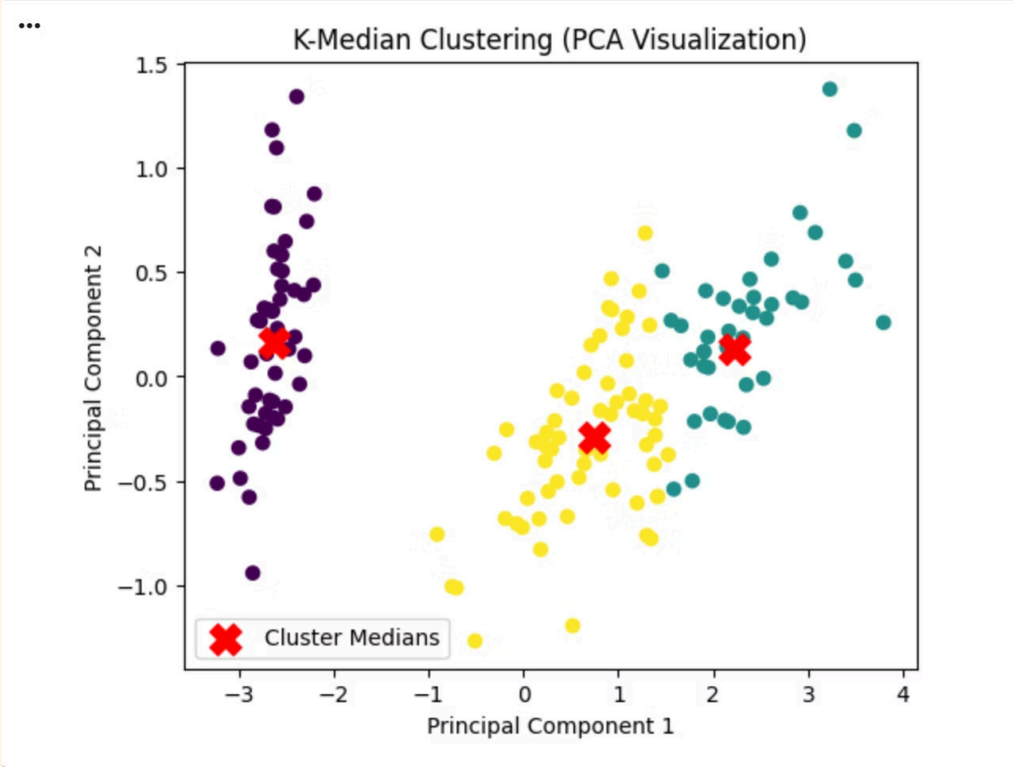
## Algorithmic Variation

K-Median replaces mean calculation with median for centroid updates.

## Robustness Advantage

Median resists outlier influence better than mean:

- Less sensitive to extreme values
- More representative of central tendency
- Stable with noisy data




## Implementation

Added post-Milestone 3 feedback to explore robustness. Applied to Iris dataset.

**Findings:** K-Median produced similar results to K-Means on Iris data, as the dataset lacks significant outliers

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# Results & Conclusion

### K = 3 Optimal

Both elbow method and silhouette analysis consistently identified three clusters as optimal for Iris dataset

### K-Means++ Stability

Enhanced initialization reduced variance across multiple runs, producing more reliable clustering

### K-Median Similarity

Robust variant performed comparably, though Iris dataset lacks outliers requiring special handling

### Reproducible Research

Complete code, methodology, and results documented for verification and extension

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## Key Takeaways

- Validation techniques (elbow, silhouette) essential for K-selection
- Initialization method significantly impacts result stability
- Dimensionality reduction aids interpretability
- Algorithm variants address specific data characteristics

**Questions & Discussion**

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